

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Cilium Assembly (GO:0060271)	-0.07760309	197	1.877e-04	5.039e-01	CEP126:1 DYNC2H1:5 KIAA0753:28 GORAB:36 TEK1:92 DNAH5:138 FBF1:147
Replication Fork Processing (GO:0031297)	-0.18265829	37	1.220e-04	0.039e-01	EME2:124 BRCA1:206 EME1:237 MUS81:419 RAD50:740 DNA2:775 WRN:1040
DNA Repair (GO:0006281)	-0.06268261	234	1.031e-03	9.225e-01	USP45:10 XRC5C:31 ATM:60 MLH1:66 PMS2:169 BRCA1:206 EME1:237 MUS81:419
G Protein–Coupled Glutamate Receptor Sig	0.32713099	9	6.790e-04	9.225e-01	TRPM1:87 HOMER3:13 GRM5:318 GRM8:415 GRM1:445 GRM4:491 GRM3:5881
Double–Strand Break Repair (GO:0060302)	-0.08361591	134	8.746e-04	9.225e-01	ZMYND8:23 XRC5C:31 ATM:60 EME1:237 RAD50:740 RNF169:132 BRCA1:206 EME1:237
Positive Regulation Of Cellular Componen	0.11602502	70	8.064e-04	9.225e-01	CLU5:5 TGFBI:22 ATP10A:56 LRP1:67 BID:143 DCTN1:267 LRP4:289
'De Novo' AMP Biosynthetic Process (GO:0	-0.10120681	4	4.833e-01	9.602e-01	PFAS:33 PAICS:6785 ATIC:6785 ADSL1:6785 NA NA
'De Novo' Post–Translational Protein Fold	0.12126361	15	1.040e-01	9.602e-01	SDF2L1:76 GAK:122 HSPAB:475 ENTDP5:5881 PTGES3:5881 DNAJB13:5881 HSPA2:5881
2–Oxoglutarate Metabolic Process (GO:000	0.05634844	13	4.819e-01	9.602e-01	OGDHL:151 D2HGDH:960 IDH1:5881 GOTT2:5881 GOTT1:5881 ADHFE1:5881
3–UTR–mediated mRNA Destabilization (GO	0.07067574	11	4.171e-01	9.602e-01	MOV10:609 DHX36:916 RC3H1:5881 ZFP361:5881 ZFP361:5881 ZFP362:5881 ZFP365:5881
3–UTR–mediated mRNA Stabilization (GO:0	0.10615460	11	2.229e-01	9.602e-01	DAZL:107 MAPK14:886 BOLL:963 ZFP36:5881 TARDBP:5881 YBX3:5881 MYD88:5881
3–Phosphoadenosine 5'–Phosphosulfate Me	-0.04012683	8	6.943e-01	9.602e-01	ENPP1:436 PAPSS2:1069 TPST2:6785 SULT2B1:6785 SULT1B1:6785 SULT1E1:6785 SULT1A3:6785
5S Class rRNA Transcription By RNA Polym	-0.05186503	5	6.880e-01	9.602e-01	PTF3C5:1509 GTF3C2:6785 GTF3C3:6785 GTF3C4:6785 GTF3C6:6785 NA NA
7–Methylguanosine RNA Capping (GO:000945	0.25805223	2	2.062e-01	9.602e-01	RNMT:188 CMTR2:5881 NA NA NA NA NA
7–Methylguanosine mRNA Capping (GO:000663	0.25805223	2	2.062e-01	9.602e-01	RNMT:188 CMTR2:5881 NA NA NA NA NA
ADP Transport (GO:0015866)	0.11126131	5	3.889e-01	9.602e-01	SLC25A41:826 SLC25A42:5881 SLC25A24:5881 SLC25A5:5881 SLC25A23:5881 NA NA
AMP Biosynthetic Process (GO:0006167)	-0.13442773	6	2.542e-01	9.602e-01	PFAS:33 APRT:864 PAICS:6785 ATIC:6785 ADSS1:6785 NUDT2:6785 NA
AMP Metabolic Process (GO:0046033)	0.07641078	9	4.274e-01	9.602e-01	NTSE:365 AK9:1217 ADSS1:5881 NUDT2:5881 NT5C1A:5881 AMPD1:5881 AMPD2:5881
ATF6–mediated Unfolded Protein Response	-0.05345055	5	7.898e-01	9.602e-01	DDIT3:1400 MBTPS1:6785 XBP1:6785 ATF6B:6785 CREBZF:6785 NA NA
ATP Metabolic Process (GO:0046034)	-0.03454527	25	5.501e-01	9.602e-01	TGFB1:22 MYH7:163 NTSE:365 HSPAB:475 NUDT2:5881 ATP5O:5881 SLC25A13:5881
ATP Synthase Coupled Electron Transport	-0.11553597	3	8.826e-01	9.602e-01	NDUFV3:1179 NDUF6:6785 NDUF2S:6785 NA NA NA
ATP Transport (GO:0015867)	0.02195108	9	6.196e-01	9.602e-01	SLC25A41:826 SLC25A42:5881 SLC25A24:5881 SLC25A5:5881 SLC25A23:5881 CALHM3:5881 SLC17A9:5881
AV Node Cell Action Potential (GO:008601)	0.07535371	6	3.617e-01	9.602e-01	SCN5A:124 CACNB2:5881 SCN5A:5881 SCN4B:5881 CACNA1G:5881 TRPM4:5881 NA
AV Node Cell To Bundle Of His Cell Commu	0.03089431	4	3.306e-01	9.602e-01	SCN5A:5881 CXADR:5881 GJC3:5881 GJA3:5881 NA NA
Arp2/3 Complex–Mediated Actin Nucleation	-0.19273439	10	3.845e-02	9.602e-01	WASHC1:500 ARPC5:662 WHAMM:1033 ARPCSL1:1228 TRIM27:1940 ARPC1A:2019 ARPC3:6785
B Cell Activation (GO:0042113)	0.05020335	61	1.757e-01	9.602e-01	HDAC9:183 HDAC4:242 CD86:253 CD40:389 DCAF1:400 FLT3:506 TPDS2:629
B Cell Activation Involved In Immune Res	0.03091945	12	7.108e-01	9.602e-01	MSH2:5881 GPR183:5881 CD40L:5881 CML1:5881 RAB8A:681 UG4:5881 RNF8:5881 RNF168:5881
B Cell Chemotaxis (GO:0035754)	-0.04914117	5	0.306e-01	9.602e-01	GAS6:1685 HKDC4:6785 HSD3B7:6785 CXCL13:6785 CYP7B1:6785 IL4:1022
B Cell Differentiation (GO:0030183)	-0.04313141	44	3.435e-01	9.602e-01	HDAC9:183 HDAC4:242 DCAF1:400 FLT3:506 CD86:253 TUBB4A:591 TRIM27:629
B Cell Mediated Immunity (GO:0019724)	-0.03123557	13	9.666e-01	9.602e-01	SLA2:1193 MPL:1200 IL9R:1688 CD70:6785 MSH2:6785 FCGR2B:6785 CD19:6785
B Cell Proliferation (GO:0042100)	0.08607861	15	2.485e-01	9.602e-01	CD40:389 MEFC2:979 IL10:5881 CD70:5881 IFCN:5881 PTPRC:5881 IFNK:5881
B Cell Receptor Signaling Pathway (GO:00	0.09317267	26	1.003e-01	9.602e-01	CD38:283 CTLA4:625 VAV3:651 MEFC2:979 TCE1:1073 SH2B2:1198 PTPRC:5881
BMP Signaling Pathway (GO:0030509)	-0.06737527	42	1.291e-01	9.602e-01	SLC39A5:455 LEFTY1:595 RUNX2:710 GDF9:795 INHCB:613 INHCB:613 INHCB:613
C–terminal Protein Amino Acid Modificati	-0.20064384	8	4.919e-02	9.602e-01	GPLD1:188 AGBL5:650 AGBL4:896 FOLH1:1730 AGTPBP1:6785 ATG16L1:6785 ATG7:6785
C–terminal Protein Deglutamylation (GO:0	-0.30359691	4	3.550e-02	9.602e-01	AGBL5:650 AGBL4:896 FOLH1:1730 AGTPBP1:6785 NA NA NA
C–terminal Protein Lipidation (GO:0006050	-0.14089015	3	3.980e-01	9.602e-01	GPLD1:188 ATG16L1:6785 ATG7:6785 NA NA NA
C4–dicarboxylate Transport (GO:0015740)	0.03726866	10	6.833e-01	9.602e-01	SLC25A12:46 SLC25A18:569 SLC25A13:5881 SLC13A2:5881 SLC1A1:5881 UCP2:5881 LRRRC8:5881
CD4–positive, Alpha–Beta T Cell Activati	0.03089745	5	8.109e-01	9.602e-01	STOML2:5881 TNFSF4:5881 RASD2:5881 NKG7:5881 HMG81:5881 NA NA
CD40 Signaling Pathway (GO:0023035)	0.03937031	7	7.183e-01	9.602e-01	CD86:253 CD40LG:5881 PHB2:5881 TNIP2:5881 TNIP2:5881 TRF1:5881 ITGB1:10857
CENP–A Containing Chromatin Assembly (GO	-0.05491412	5	6.707e-01	9.602e-01	HJURP:1304 MIS18A:6785 CENPI:6785 OIP5:6785 NASP:6785 NA NA

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
NIKOLSKY_BREAST_CANCER_1Q21_AMPLICON	-0.26593169	27	1.741e-06	1.128e-02	PBXIP1:32 ADAM15:55 RUSC1:113 MTX1:253 DCST2:335 THBS3:483 SHC1:558
MIKKELSEN_ES_ICP_WITH_H3K4ME3	-0.05364367	532	2.757e-05	3.780e-02	SLC2A10:19 ZMYND23:23 PBXIP1:32 ZBTB4:42 ANXA3:48 WDCP:72 FAM149A:101
REACTOME_DNA_DOUBLE_STRAND_BREAK_REPAIR	-0.11504674	111	2.918e-05	3.780e-02	XRC5C:31 ATM:60 EME2:124 BRCA1:206 EME1:237 MUS81:419
REACTOME_MEIOTIC_RECOMBINATION	-0.22184976	31	1.926e-05	3.780e-02	ATM:60 MLH1:66 PMS2:169 BRCA1:206 RAD50:740 H2AX:792 RPA1:904
REACTOME_DISEASES_OF_DNA_REPAIR	-0.17815757	47	2.413e-05	3.780e-02	ATM:60 MLH1:66 PMS2:169 BRCA1:206 MUTHYH:211 HUS1:459 MSK1:509
REACTOME_DEVELOPMENTAL_BIOLOGY	0.04592905	669	6.278e-05	5.816e-02	CACNA1S:17 TGFBI:22 CNTN1:26 CDK8:31 PPARA:34 DEK:54 KLM5:58
REACTOME_CYTOKINE_SIGNALING_IN_IMMUNE_SY	0.05420268	466	7.010e-05	5.816e-02	COL1A2:9 TGFBI:22 IL33:24 TNFRSF4:39 RELB:47 IL1RL2:121 TRIM68:137
REACTOME_RESOLUTION_OF_D_LOOP_STRUCTURES	-0.19802201	33	8.314e-05	5.816e-02	ATM:60 EME2:124 BRCA1:206 EME1:237 MUS81:419 RAD50:740 DNA2:775
WP_DNA_REPAIR_PATHWAYS_FULL_NETWORK	-0.11467024	98	8.981e-05	5.816e-02	XRC5C:31 ATM:60 MLH1:66 PMS2:169 BRCA1:206 MUTHYH:211 FANCE:420
WP_CILIOPATHIES	-0.09401272	148	8.217e-05	5.816e-02	DYNC2H1:5 KIAA0753:28 DDCD2:192 SDCCAG8:200 DNAH11:217 CEP164:233 PKD1L1:316
MIKKELSEN_NPC_ICP_WITH_H3K4ME3	-0.06393525	310	1.180e-04	6.371e-02	ZMYND8:23 PBXIP1:32 WGDG:72 KDM6B:180 ZNF608:216 CAVIN2:244 CSRPNP3:286
REACTOME_HDR_THROUGH_HOMOLOGOUS_RECOMBIN	-0.14677250	58	1.121e-04	6.371e-02	ATM:60 EME2:124 BRCA1:206 EME1:237 MUS81:419 HUS1:459 RFC5:530
REACTOME_HDR_THROUGH_SINGLE_STRAND_ANNIEA	-0.19290778	32	1.599e-04	7.504e-02	ATM:60 BRCA1:206 HUS1:459 RFC5:530 RAD50:740 DNA2:775 ABL1:797
KIM_ALL_DISORDERS_CALB1_CORR_UP	0.06075307	330	1.622e-04	7.504e-02	SLC25A12:46 SLC20A1:49 NCKAP1:102 CHGB:117 ATP1A3:146 TUBB1:192 DCM2A:247
REACTOME_INFECTIOUS_DISEASE	0.04764474	539	1.789e-04	7.722e-02	TLR8:10 ENTDP1:18 TGFBI:22 SH3GL3:38 AP1B1:44 NPTN:459 TRIM27:629
WP_22Q112_COPY_NUMBER_VARIATION_SYNDROME	0.12260096	77	2.032e-04	8.132e-02	HIRA:3 DGC9:52 DEPDC5:60 ZDHHC8:130 PIKA:231 SERPIND1:250 KLHL22:268
KAUFFMANN_DNA_REPAIR_GENES	-0.08003677	181	2.135e-04	8.132e-02	XRC5C:31 ATM:60 MLH1:66 MLH3:90 PMS2:169 BRCA1:206 MUTHYH:211
REACTOME_HOMOLOGY_DIRECTED_REPAIR	-0.11198819	89	2.659e-04	9.566e-02	ATM:60 EME2:124 BRCA1:206 EME1:237 MUS81:419 HUS1:459 RFC5:530
REACTOME_HOMOLOGY_DNA_PAIRING_AND_STRAI	-0.15910278	39	8.896e-04	1.909e-01	ATM:60 BRCA1:206 HUS1:459 RFC5:530 RAD50:740 DNA2:775 RPA1:904
REACTOME_DNA_REPAIR	-0.06871598	226	5.768e-04	1.909e-01	USP45:10 XRC5C:31 ATM:60 MLH1:66 EME1:237 PMS2:169 BRCA1:206
REACTOME_RNA_POLYMERASE_II_TRANSCRIPTION	0.03884155	683	6.315e-04	1.947e-01	IWS1:17 COL1A2:9 TGFBI:22 MYEK:23 CDK8:31 PPARA:34 DEK:54
WP_INTERACTIONS_BETWEEN_IMMUNE_CELLS_AND	0.20332916	23	7.390e-04	2.175e-01	TLR8:10 TGFBI:22 CD80:210 TGFBR2:234 CD86:253 CTLA4:625 TLR7:649
WP_JOUBERT_SYNDROME	-0.12076775	64	8.438e-04	2.376e-01	ATM:60 CEP164:233 ARL13B:363 MKS1:438 RAB8A:681 TCTN2:693 RAD50:740
REACTOME_RESOLUTION_OF_D_LOOP_STRUCTURES	-0.18559439	26	1.058e-03	2.854e-01	ATM:60 BRCA1:206 RAD50:740 DNA2:775 BRP1:1030 WRN:1040 BLM:1348
WINTER_HYPOXIA_METAGENE	0.06982455	176	1.443e-03	3.738e-01	TGFBI:22 PPARA:34 SLC20A1:49 ENG:51 CD86:253 TUBB4A:591 MGRN1:138
CONRAD_STEM_CELL	0.15142172	27	1.640e-03	4.000e-01	CD9:7 DNMT3B:140 MYC:187 PWIL2:151 DKK3:305 RET:321 MYBL2:396
WP_T_CELL_MODULATION_IN_PANCREATIC_CANCE	0.17518380	36	1.668e-03	4.000e-01	ENTPD1:18 TGFBI:22 TNFRSF4:39 CD80:210 CD86:253 NTSE:365 CD40:389
WP_NETWORK_MAP_OF_SARSOCV2_SIGNALING_PAT	0.07281098	156	1.745e-03	4.035e-01	IL33:24 TNFSF10:78 RRM2:89 CD163:114 BID:143 ITIH3:171 TRAF2:202
REACTOME_ADAPTIVE_IMMUNE_SYSTEM	0.04240996	463	1.927e-03	4.160e-01	COL1A2:9 NFKBIE:32 HERC2:37 AP1B1:44 UBR2:50 TUBB4A:59 MGRN1:138
REACTOME_MEIOSIS	-0.12222087	54	1.907e-03	4.160e-01	ATM:60 MLH1:66 MLH3:90 BRCA1:206 STAG1:281 RAD50:740 H2AX:792
KEGG_CYTOKINE_CYTOKINE_RECEPTOR_INTERACT	0.06697460	190	2.003e-03	4.184e-01	TGFBI:22 TNFRSF4:39 TNFSF10:78 TGFBR2:234 TNFSF10:78 TNF:348 TNFSF14:350
REACTOME_LOSS_OF_FUNCTION_OF_SMAPD2_3_IN	0.39425241	5	2.265e-03	4.314e-01	TGFBI:22 ZFYVE9:86 TGFBR2:234 SMAD2:481 SMAD4:5881 NA NA
REACTOME_SIGNALING_BY_TGF_BETA_RECEPTOR	0.39425241	5	2.265e-03	4.314e-01	TGFBI:22 ZFYVE9:86 TGFBR2:234 SMAD2:481 SMAD4:5881 NA NA
WP_SMC1SMC3_ROLE_IN_DNA_DAMAGE_CORNELIA	-0.31272267	8	2.192e-03	4.314e-01	ATM:60 BRCA1:206 RAD50:740 PAXIP1:1061 MRE11:1398 RAD18:2001 SMC2A1:6785
REACTOME_SIGNALING_BY_INTERLEUKINS	0.05035864	312	2.349e-03	4.347e-01	COL1A2:9 TGFBI:22 IL33:24 IL1RL2:121 MYC:187 TRAF2:202 CD80:210
REACTOME_TRANSCRIPTIONAL_REGULATION_BY_T	0.17889236	24	2.423e-03	4.358e-01	DEK:54 YY1:160 MYC:187 MYBL2:396 TRAF2:202 TRAF2:202 TRAF2:202
PID_FANCONI_PATHWAY	-0.13271960	43	2.617e-03	4.581e-01	ATM:60 BRCA1:206 FANCE:420 HUS1:459 RFC5:530 RAD50:740 H2AX:792
REACTOME_RNA_POLYMERASE_II_TRANSCRIPTION	0.14659116	35	2.701e-03	4.603e-01	SYMPK:23 SLU17:39 THOCS1:98 LSM11:285 SLBP:409 CPSPF1:443 SRSF9:568
MIKHAYLOVA_OXIDATIVE_STRESS_RESPONSE_VIA	0.24563137	4	3.196e-03	4.813e-01	QAT:694 CTSD:761 AKR1B1:1136 PGAM1:1180 NA NA NA
KEGG_NON_HOMOLOGOUS_END_JOINING	-0.25685973	11	3.169e-03	4.813e-01	XRC5C:31 RAD50:740 DNTT:1012 POLL:1047 XRC4:1239 NHEJ1:1351 MRE11:1398

DisGeNET Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Hepatitis C	0.06750325	498	3.611e-07	3.526e-03	CLU5:5 TLR8:10 TGFBI:22 PPARA:34 ACE:36 DEPDC5:60 TNFSF10:78
Encephalomyelitis	0.08513526	254	3.471e-06	1.695e-02	CLU5:5 TLR8:10 ENTDP1:18 TGFBI:22 PPARA:34 ACE:36 TNFRSF4:39
Dermatitis, Allergic Contact	0.13501732	83	2.202e-05	4.300e-02	PPARA:34 ACE:36 CNTN3:179 CD80:210 MRC1:1224 HSD11B1:344 TNF:348
HIV Infections	0.05582680	511	2.045e-05	4.300e-02	TLR8:10 ENTDP1:18 TGFBI:22 GLUL:28 TRPV1:45 TGFBI:22
Psoriasis	0.05514324	522	2.134e-05	4.300e-02	TLR8:10 HCA2R:19 TGFBI:22 IL33:24 PPARA:34 ACE:36 AQP1:142
Aortic Aneurysm, Abdominal	0.08580122	196	3.780e-05	6.152e-02	COL1A2:9 TGFBI:22 PPARA:34 ACE:36 FCGRT:63 LRP1:67 OPN1LW:105
Seminoma	0.09992323	135	6.476e-05	9.035e-02	CLU5:5 ACE:36 ENTDP6:99 AFP:118 DNMT3B:140 GCTL3:177 HDAC9:183
Dermatitis	0.09661719	138	9.363e-05	1.046e-01	TGFBI:22 IL33:24 PPARA:34 TRPA1:140 KLTK:74 LPN2:110 PTGDS:164
Inflammatory dermatitis	0.10557842	115	9.636e-05	1.046e-01	TGFBI:22 IL33:24 PPARA:34 TRPA1:140 LPN2:110 PTGDS:164 ITGA7:274
Myasthenia Gravis	0.10246538	120	1.112e-04	1.088e-01	TGFBI:22 TNFRSF4:39 DNMT3B:140 CD86:253 ITGA7:274 KRT5:274
Scleroderma	0.10185243	119	1.302e-04	1.155e-01	COL1A2:9 TGFBI:22 ENG:51 RTCA:34 YY1:160 COMP:169 MYC:187
Dermatitis, Atopic	0.06237939	316	1.563e-04	1.174e-01	TGFBI:22 IL33:24 PPARA:34 RELB:47 KLTK:74 FCGRT:63 KLTK:74
Eczema	0.06378004	306	1.556e-04	1.174e-01	TGFBI:22 IL33:24 PPARA:34 KLTK:55 KLTK:74 TRPM1:67 EMSY:119
Malignant neoplasm of stomach	0.03068705	1450	1.801e-04	1.256e-01	CLU5:5 COL1A2:9 TGFBI:22 SYMPK:23 IL33:24 CNTN1:26 GLUL:28
Lymphoma, T–Cell, Cutaneous	0.08092886	177	2.182e-04	1.421e-01	TGFBI:22 TNFSF10:78 SCCEP1:128 HDAC9:183 MYC:187 TRAF2:202 CD80:210
Hepatocarcinogenesis	0.05865038	326	3.077e-04	1.878e-01	CLU5:5 TGFBI:22 GLUL:28 PPARA:34 TRIM24:57 TNFSF10:78 AFP:118
Epilepsy, Temporal Lobe	0.09330872	122	3.874e-04	2.033e-01	GLUL:28 ACE:36 TRPV1:45 DEPDC5:60 TNFSF10:78 SCNA1A:145 GRM5:318
Chronic Lymphocytic Leukemia	0.03985666	708	3.954e-04	2.033e-01	ENTPD1:18 TGFBI:22 NFKBIE:32 PPARA:34 ACE:36 AQP1:142 RELB:47
Pemphigus Vulgaris	0.15896573	42	3.692e-04	2.033e-01	FCGRT:63 GJA8:120 DSCG3:213 TN